Hamiltonian Squares in Directed Graphs

Semi-degree conditions and oriented discrepancy

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Outline

1. Two questions

2. Semi-degree condition

3. Oriented discrepancy

4. Open directions

Two questions on Hamilton squares in digraphs

Question 1: Existence in digraphs

How large must the minimum semi-degree be to force a *directed* square of Hamilton cycles?

$$\delta^0(D) \quad \Longrightarrow \quad C_n^2 \subseteq D.$$

Two questions on Hamilton squares in digraphs

Question 1: Existence in digraphs

How large must the minimum semi-degree be to force a *directed* square of Hamilton cycles?

$$\delta^0(D) \quad \Longrightarrow \quad C_n^2 \subseteq D.$$

Question 2: Oriented discrepancy

In an oriented graph, can a square of Hamilton cycle be chosen with many edges pointing the same way?

$$\delta(G) \quad \Longrightarrow \quad \sigma_{\max}(H) \text{ large.}$$

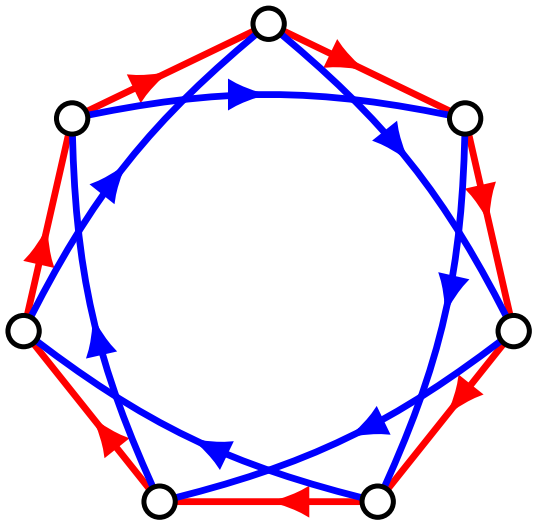
k-th power of a Hamilton cycle

Definition

If $C = v_1 \cdots v_n v_1$ is a directed Hamilton cycle, then

$$E(C^2) = \{v_i v_{i+1},\ v_i v_{i+2} : i \in [n]\},$$

with indices modulo n.



cycle arcs

distance-two arcs

The Pósa–Seymour threshold and its directed analogue

Theorem (Komlós, Sárközy, Szemerédi, Ann. Comb., 1998)

For fixed k and all sufficiently large n,

$$\delta(G) \geq \frac{k}{k+1}n \quad \Longrightarrow \quad C_n^k \subseteq G.$$

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Problem (DeBiasio et al., Combin. Probab. Comput., 2026)

Does the directed analogue hold?

$$\delta^0(D) \geq \frac{k}{k+1}n \quad \stackrel{?}{\Longrightarrow} \quad C_n^k \subseteq D.$$

A first semi-degree result

Theorem (Wang, Wei, Yan, 2026+)

For every $\gamma > 0$ and all sufficiently large n,

$$\delta^0(D) \geq \left(\frac{3}{4} + \gamma\right) n \quad \Longrightarrow \quad C_n^2 \subseteq D.$$

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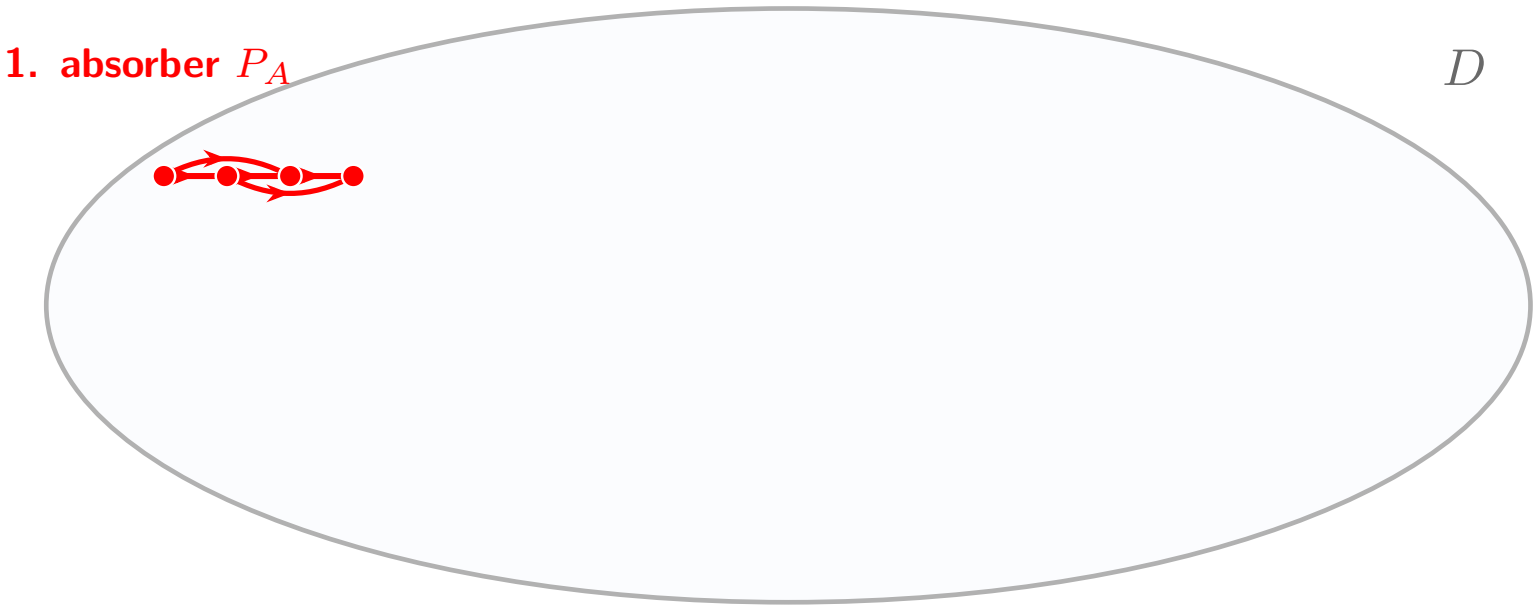
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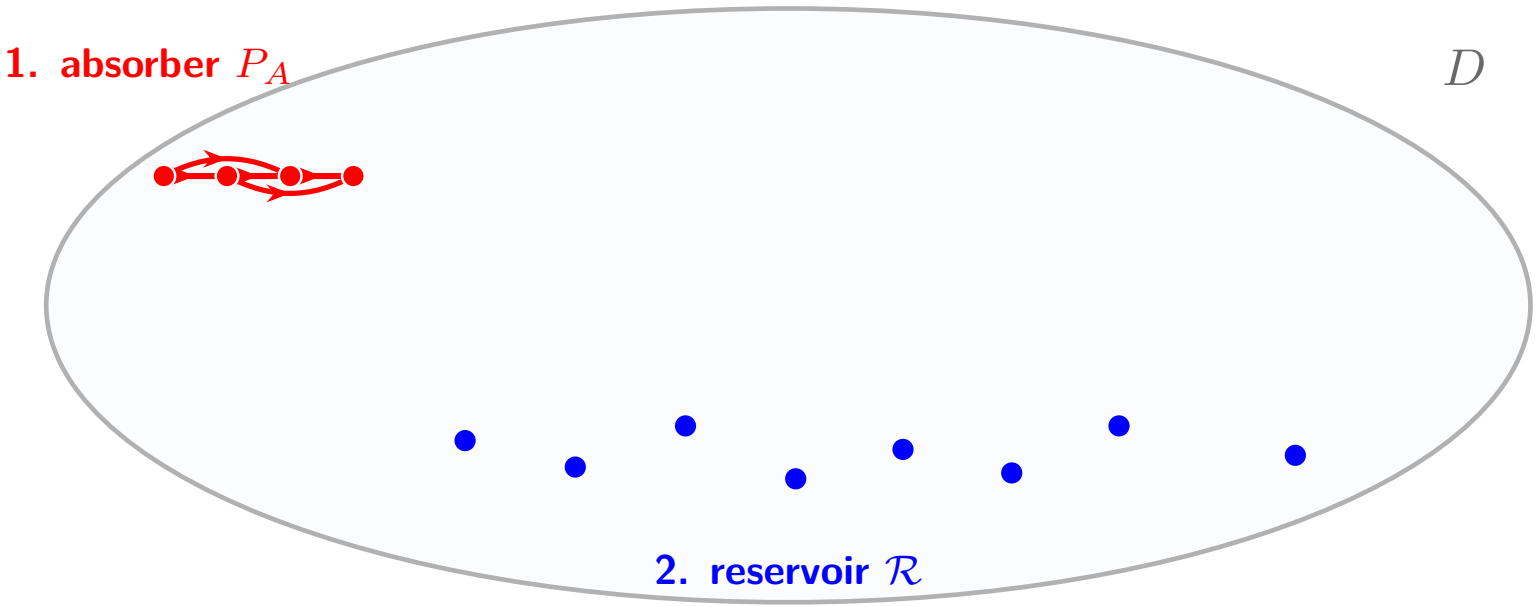


Factor consequence. On ℓm vertices: a C_ℓ^2-factor for large ℓ.

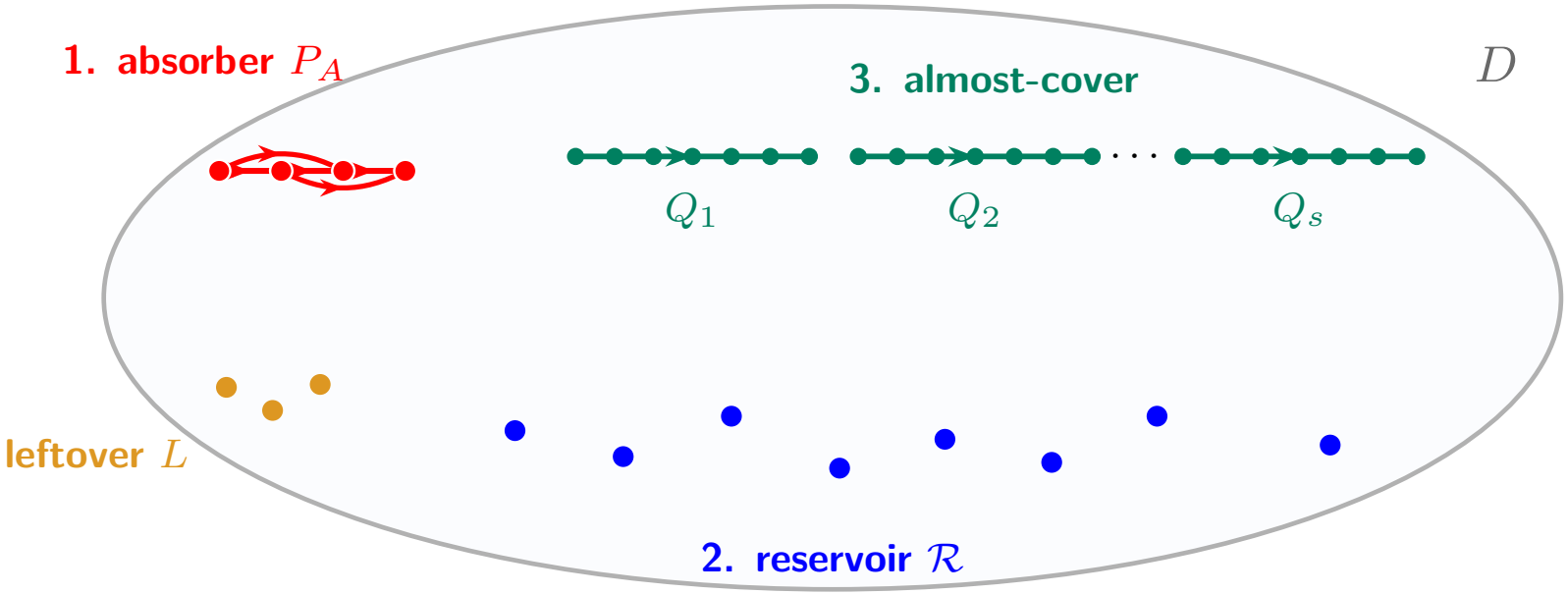
Step 1: choose many absorbers



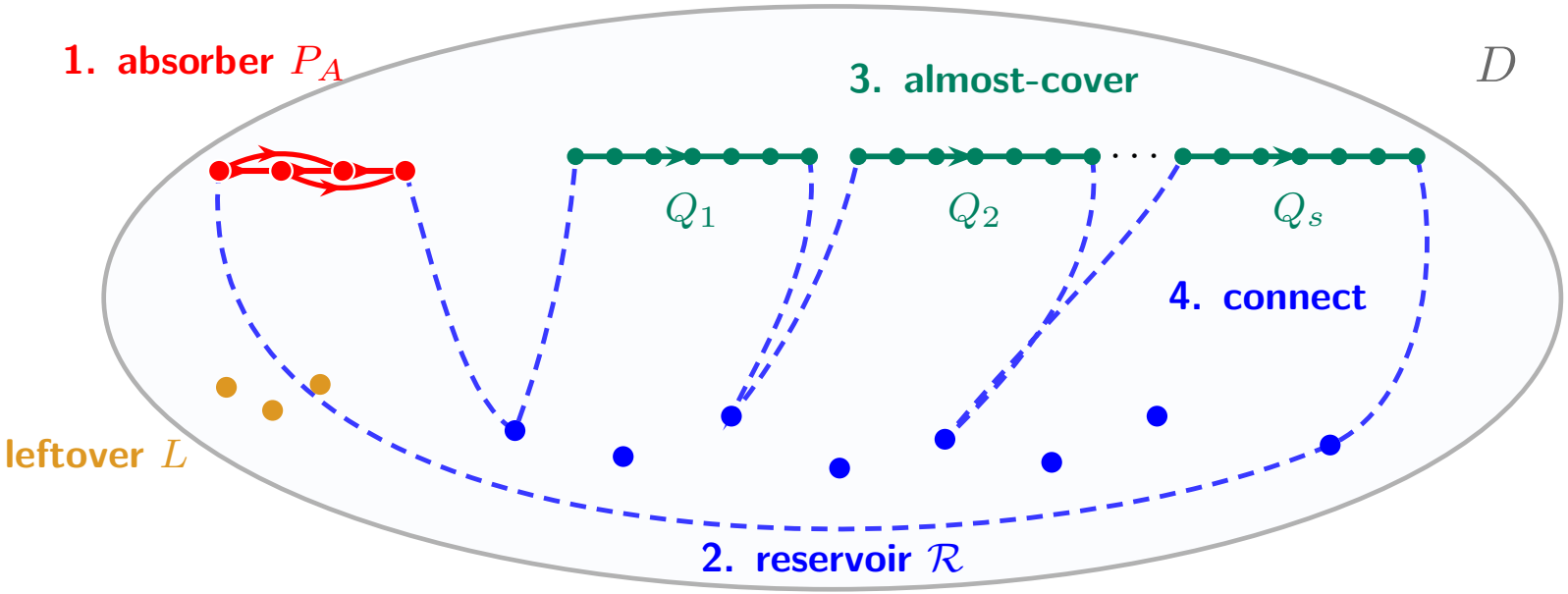
Step 2: choose a reservoir



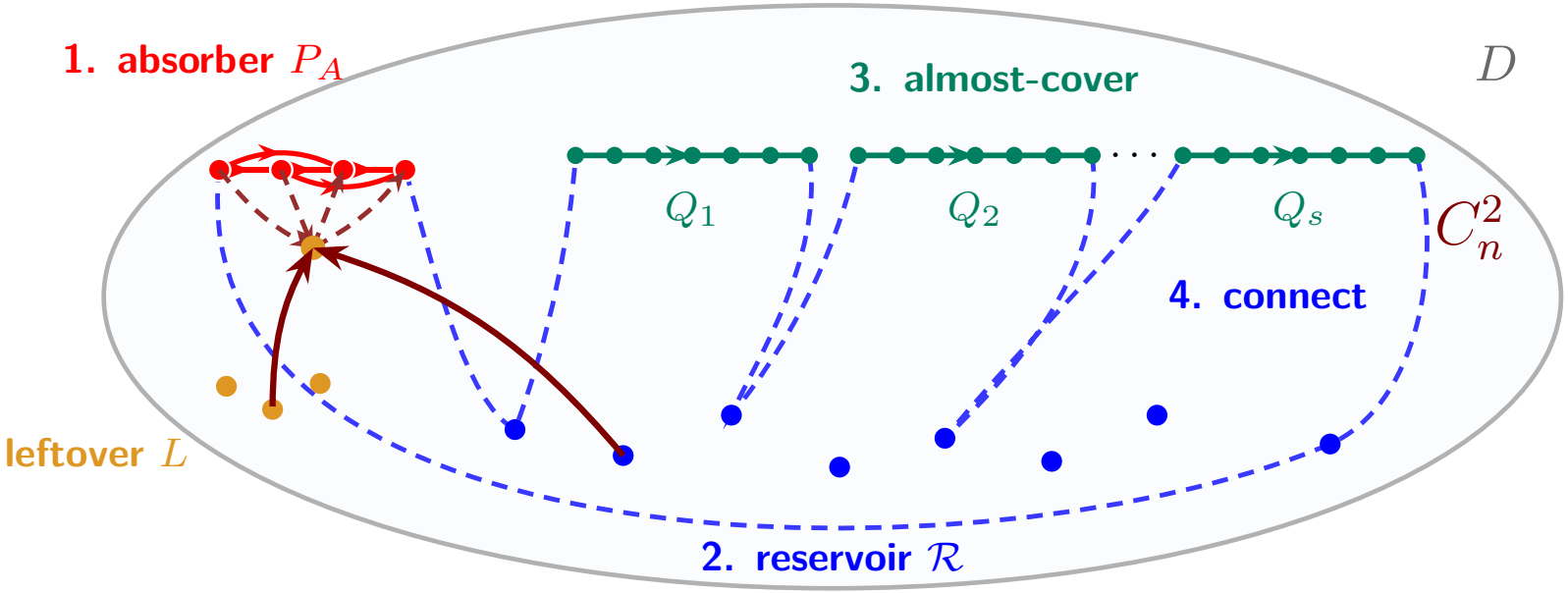
Step 3: cover almost all remaining vertices



Step 4: connect the paths cyclically



Step 5: absorb the leftover



Recent progress

Theorem (Wang, Wei, Yan, arXiv:2607.13831)

We settle the total degree conjecture of DeBiasio et al. for $k = 2$.

$$\delta(D) \geq \frac{8}{5}n - c(n) \quad \Longrightarrow \quad C_n^2 \subseteq D.$$

Here $\delta(D) = \min_v(d_D^+(v) + d_D^-(v))$, and $c(n) = 2$ if $n \equiv 2, 4 \pmod 5$, while $c(n) = 1$ otherwise.

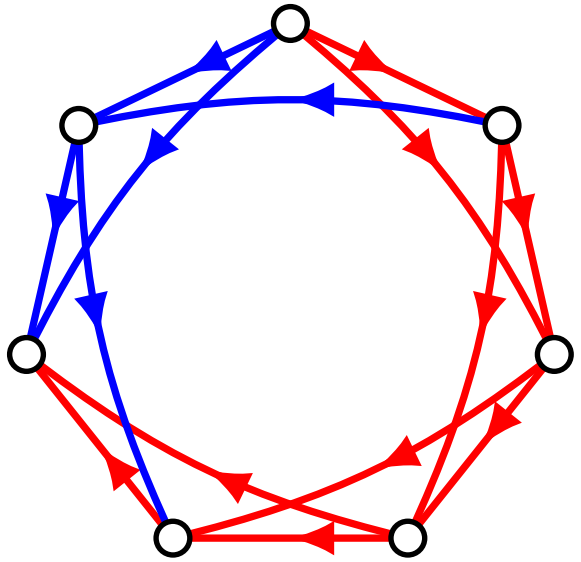
Oriented discrepancy

Definition

Let $C = v_1v_2\cdots v_nv_1$ be a Hamilton cycle such that $H := C^2$ lies in the underlying graph of an oriented graph G.

- $\sigma^+(H) := \#\{(i,j) : v_iv_{i+j} \in E(G)\}$ **(forward edges)**
- $\sigma^-(H) := \#\{(i,j) : v_{i+j}v_i \in E(G)\}$ **(backward edges)**
- $\sigma_{\max}(H) := \max\{\sigma^+(H), \sigma^-(H)\}$

Here $i \in [n]$, $j \in \{1,2\}$, and indices are taken modulo n.



Motivation

Theorem (Freschi, Lo, J. Combin. Theory Ser. B, 2024)

If $\delta(G) \geq n/2$, then G contains a Hamilton cycle C with $\sigma_{\max}(C) \geq \delta(G)$.

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Problem (Freschi, Lo, J. Combin. Theory Ser. B, 2024)

Determine the largest lower bound on $\sigma_{\max}(H)$ that holds for k-th power of Hamilton cycles $H \subseteq G$ under minimum degree condition.

The main discrepancy theorem

Theorem (Chang, Cheng, Wang, Wei, Yan, 2026+)

For every $\xi > 0$, every sufficiently large oriented graph G with

$$\delta(G) \geq \frac{2}{3}n$$

contains a square of Hamilton cycle H satisfying

$$\sigma_{\max}(H) \geq \max\{n,\ f(\delta(G), n) - \xi n\}.$$

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Corollary

If G contains no tournament on six vertices, then

$$\sigma_{\max}(H) \geq \max\{n,\ 3\delta(G) - n - \xi n\}.$$

The function $f(\delta(G), n)$

Degree range (Balogh–Freschi–Treglown, *EJC*, 2025). Choose r with

$$\left(1 - \frac{1}{r-1}\right) n \leq \delta(G) \leq \left(1 - \frac{1}{r}\right) n.$$

Mixed-tiling weights.

$$A_r = (r-1)\delta(G) - (r-2)n, \qquad \overline{A}_r = (r-1)n - r\delta(G).$$

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Cover contribution.

$$\boxed{f(\delta(G), n) = A_r N_r + \overline{A}_r N_{r-1}}$$

Near $2n/3$: $f(\delta(G), n) = 3\delta(G) - n$.

Bias from the almost-covering step

directed regularity → **reduced oriented graph**

Minimum degree passes to the reduced graph.

Bias from the almost-covering step

directed regularity → **reduced oriented graph** → T_r, T_{r-1} **tiling** → **biased 2-path cover**

Lift the tiling to $\mathcal{Q}$ with

$$\sigma_{\max}(\mathcal{Q}) \geq f(\delta(G), n) - o(n).$$

Bias from the almost-covering step

directed regularity → **reduced oriented graph** → T_r, T_{r-1} **tiling** → **biased** 2**-path cover** → **connect + absorb Hamilton square**

Connect and absorb using $o(n)$ vertices.

The total loss is at most ξn.

Three open directions remain

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1. Exact semi-degree threshold: $2n/3$?
2. Optimal discrepancy: determine N_r (or $M_r = 2r - N_r$), $r \geq 6$.
3. Higher powers $k \geq 3$ and Ore-type conditions.

Thanks!